

Maven Lifecycle - Simple Practical Guide

- 1. Validate:** Checks project structure and pom.xml correctness.
- 2. Compile:** Compiles source code into .class files.
- 3. Test:** Runs unit tests (JUnit/TestNG).
- 4. Package:** Creates JAR/WAR file.
- 5. Verify:** Runs additional checks like integration tests.
- 6. Install:** Installs artifact into local repository (~/.m2).
- 7. Deploy:** Uploads artifact to remote repository (Nexus/Artifactory).

Lifecycle Flow:

validate → compile → test → package → verify → install → deploy

Dependency Scope:

Scope	Description
compile	Available everywhere
provided	Available at compile time, not packaged
runtime	Needed only at runtime
test	Used only for testing